Technical record: boundaries, ground and limits

Version V4. Planar UTM19S figures are distinguished from geodesic machine metrics. Screening data is preliminary; image review is not engineering certification.

Test / capacity	Value / method
whole_site_km2	1657.8435417846936
master_km2	8.635
town_km2	5.76
dwellings	7872
residents	19680.0
jobs	2150
housing_blocks	99
school_places	3100
building_fixed_overlap_m2	0.0
building_channel_screen_overlap_m2	0.0
building_flood_screen_overlap_m2	0.0
base_gap_m2	0.0
base_overlap_m2	5.0067901611328125e-06
land_partition_gap_m2	0.0

Exclusive territorial account

Gross area and current district edge logic; fixed context is unchanged and is not counted again as new development.

District / primary use	Area km2	Edge / basis
D00-F / Fixed agrivoltaic pilot / fixed_context	13.892	Exact issued boundary; forage and jojoba retained; no redesign or new capacity credited.
D00-L / Fixed living pilot / fixed_context	2.481	Exact issued living outline; no redesign.
D07 / Mountain stewardship / ecology	269.723	Retained source-DEM steep/high terrain interpretation; not reclassified as energy to meet a target.
D06 / Drainage and dry landscape links / reserve	77.228	Retained seasonal-drainage/landscape account; not a lake or statutory exclusion; detailed suitability remains unresolved.
D05-W / Western arrival and trade reserve / logistics	16.969	Western approach to retained SW public/regional spine; angular edge is an entrant service boundary, not cadastre. First gate is access-only; no customs or throughput entitlement.
D05-E / Eastern domestic transfer reserve / logistics	12.591	Compact junction of regional east-west spine and south-east branch. Domestic consolidation only; does not claim a Chilean border port or approved highway connection.
D01 / Southwest living and civic territory / mixed_use	79.995	Compact town plus future public-service catchment at SW arrival. Northeast service limit is pulled back from prior basin; excludes separate works/farm parcels. Only the existing 5.76km2 town carries population capacity.
D04 / Central processing and repair corridor / industry	48.168	A declared 3.2km-wide operating corridor along the first15.5km of the proposed central transfer route. Replaces 252km2 general industry. Width is a service-envelope choice, not a road ROW or buildable-area result; no new operating capacity credited.
D02 / Western and central agricultural territory / agriculture	501.214	Western alluvial-fan study expanded into the central valley/service catchment released from general industry. Edges follow declared field-service envelopes; soils, lawful irrigation and flood design must qualify actual fields. Not all land is irrigated.
D03-S / Southern energy and compute works / energy	15.868	Separate existing proposed compute and solar parcels south of town; land released from the old living basin remains an energy/service study, not housing capacity.
D03-N / Northern solar-energy territory / energy	188.352	Retains northern shoulder allocation served by northern route study. No cultivation credit without a separate crop/water design.
D03-E / Central and eastern energy territory / energy	431.363	Released former oversized industrial/logistics regions now form the main conditional solar-energy/service territory, outside retained mountain/drainage units. Future field subdivisions follow access/drainage studies; polygon area is neither panel area nor power capacity.

Functional balance: three linked accounts

Whitepaper V7 page5 / strategic reference, not installed area. Planar UTM19S; corrected geodesic audit is separately retained in geometry-and-balance-qa.json.

Function	Reference %	Exclusive km2 / %	Overlapping coverage km2 / %
energy_compute	60	635.58 / 38.34	1011.65 / 61.02
agri_production	30	515.11 / 31.07	515.11 / 31.07
living	5	82.48 / 4.97	82.48 / 4.97
industry	3	48.17 / 2.91	48.17 / 2.91
port	2	29.56 / 1.78	29.56 / 1.78
Retained terrain/drainage	separate	346.95 / 20.93	not energy installations

Overlap. Solar/agriculture co-use376.06km2 including fixed pilot context. This is counted once in exclusive land; functional coverage does not sum to100%.

Operating scale. Current detailed study8.635km2, town5.76km2, new crops7.656ha; no additional residents, IT MW, solar capacity or crop yield from gross territorial labels.

Future water consequence. A35% net crop/gross ratio at an assumed3,000m3/ha/year over the new362.17km2 shared-field candidate, excluding the fixed pilot, implies about38.0million m3/year of additional future water. This is neither supplied nor included in current operating demand. Resource, soil, grid, access and demand gates precede each field/campus release.

Spatial correction. Industry contracts from251.93 to48.17km2 along the transfer corridor; gateways from291.98 to29.56km2 in two compact approach/junction reservations. Most allocated land is unserviced and has no throughput or employment credit.

Complete construction capacity account

New / cumulative capacities, from current building inventory; household ratio 2.5 is assumed. Regional research districts receive no invented resident or job capacity.

Phase	Homes new / total	Residents new / total	Jobs new / total	Objective
1	1776 / 1776	4440 / 4440	905 / 905	Complete western neighbourhood, civic core, utilities and first productive halls
2	2112 / 3888	5280 / 9720	805 / 1710	Eastern living district and additional productive capacity
3	3984 / 7872	9960 / 19680	440 / 2150	Northern neighbourhoods and final staged halls

Per-zone detail is in phasing_programme.json and the phased geometry. P1 has one 750-place school for an assumed 666 pupils; P2 has 1,500 for 1,458; P3 has 3,100 for 2,952. This arithmetic does not establish service staffing, age mix, school funding, employment demand or a lawful settlement entitlement.

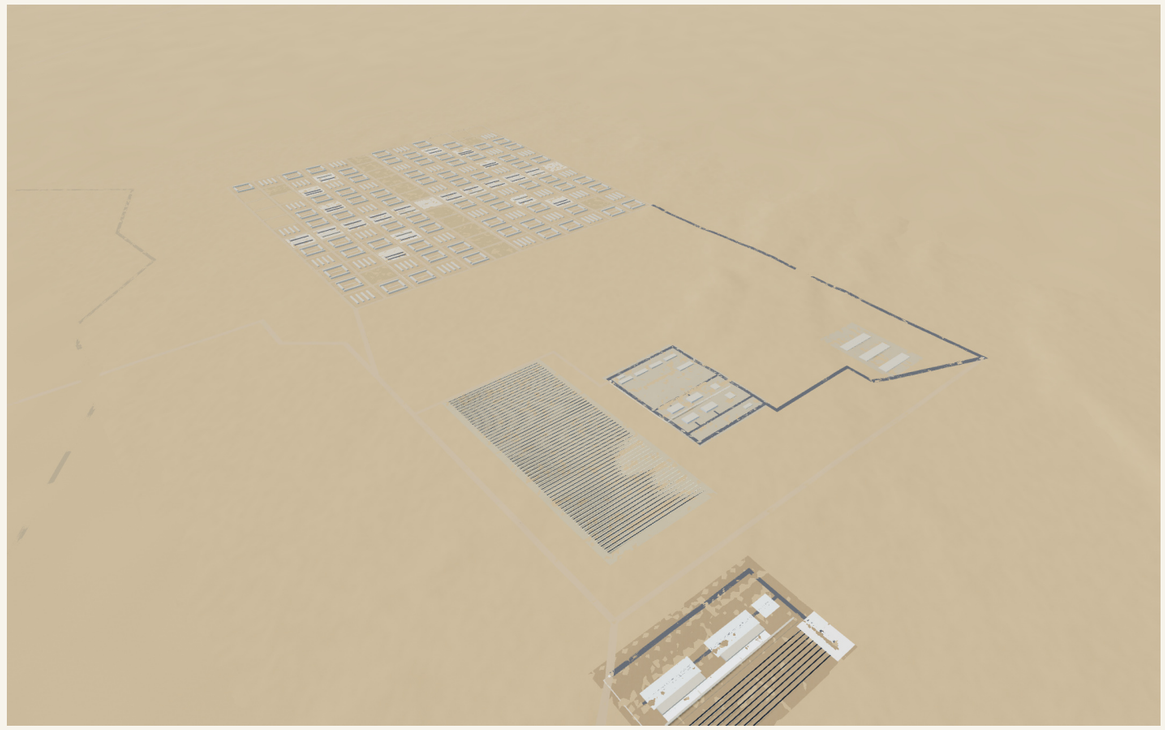
Resource scenarios: water and power

Recomputable arithmetic; new crop irrigation is included; fixed-pilot consumption remains additional and unknown.

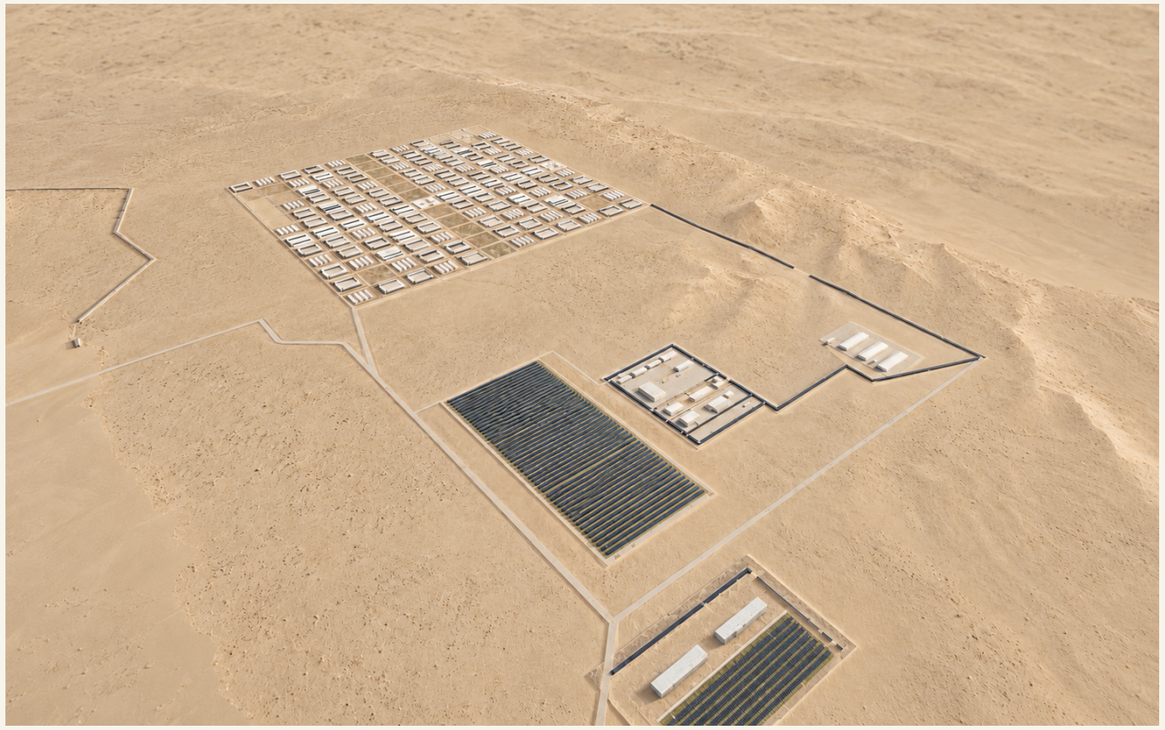
Scenario variable	Value
domestic_gross_m3y	610572.0
civic_gross_m3y	107748.0
work_gross_m3y	23575.0
compute_makeup_m3y	9110.4
gross_use_m3y	773973.4
reuse_m3y	100000
fresh_abstraction_m3y	732579.7826086957
distribution_loss_fraction	0.08
full_it_mw	8
pue_assumed	1.3
solar_mwp	26.784
solar_yield_kwh_kwp_y	1800
solar_mwh_y	48211.2
compute_mwh_y	91104.0
fixed_pilot_water	Additional and unknown; no supply entitlement confirmed
agriculture_m3y	22968
trial_irrigation	7.656ha new crop bands;22968m3/y assumed irrigation, no lawful supply or commercial yield established

city-overall / control and finished image

Complete southwest settlement system / full build-out / Full build-out scenario; node core enabled in Phase 1



ORIGINAL GEOMETRY CONTROL

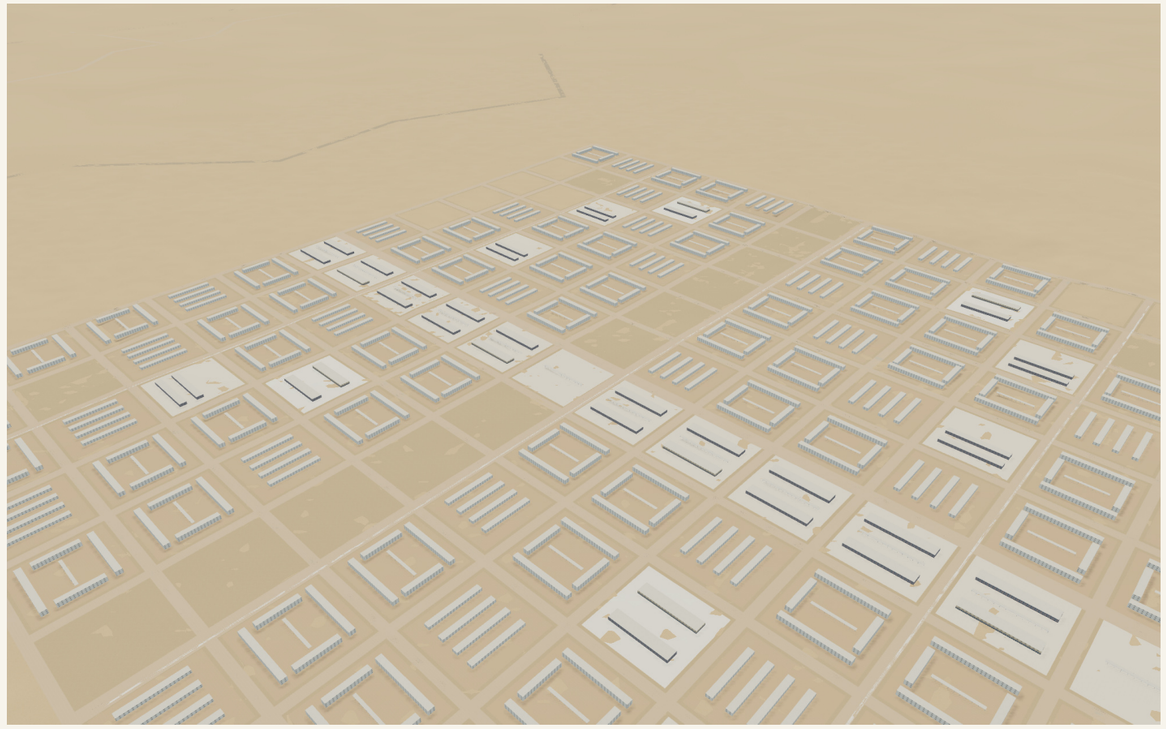


SELECTED FINISHED CONCEPT

Camera UTM19S: [572100, 6550500, 2300]; target [574100, 6553000, 0]; FOV 52 degrees. Terrain and cameras come from the current model. Maximum-corner conceptual building plinths prevent terrain clipping and require civil design. AI materials, small people, furniture and mechanical details are illustrative; exact geometry fidelity needs continued independent review.

city-centre / control and finished image

Town civic band and living districts / local aerial / Full build-out scenario; node core enabled in Phase 1



ORIGINAL GEOMETRY CONTROL

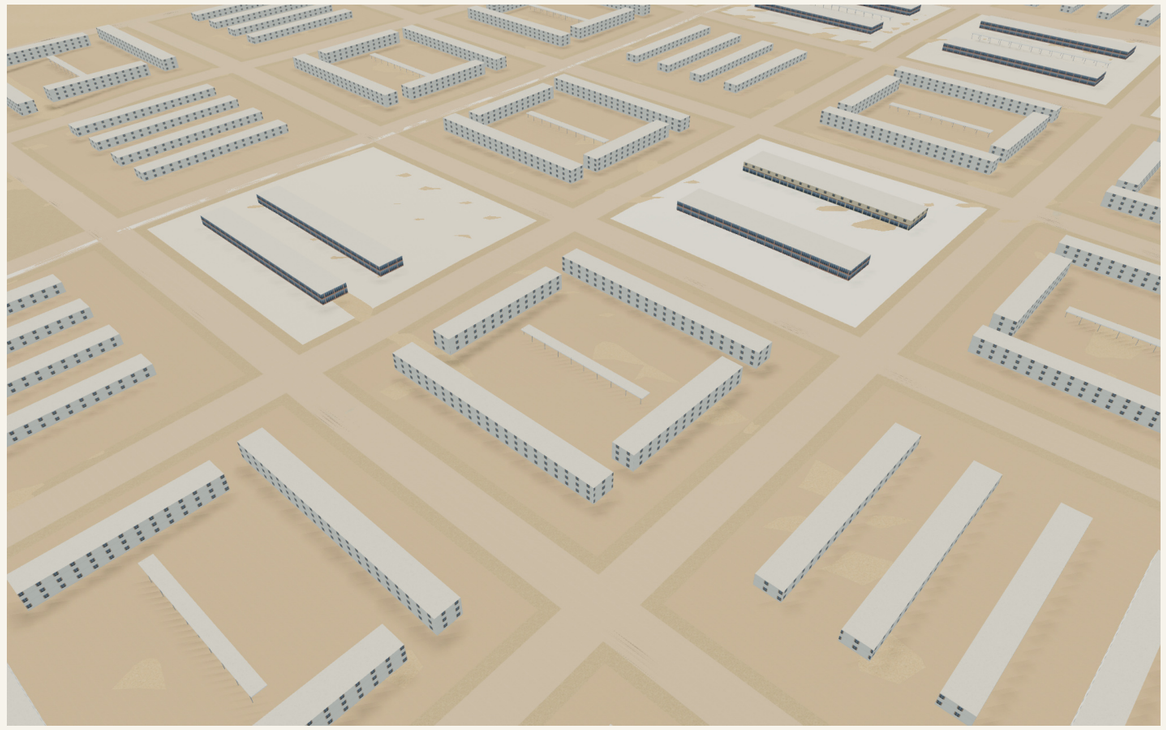


SELECTED FINISHED CONCEPT

Camera UTM19S: [575600, 6554300, 1050]; target [574500, 6555470, 0]; FOV 48 degrees. Terrain and cameras come from the current model. Maximum-corner conceptual building plinths prevent terrain clipping and require civil design. AI materials, small people, furniture and mechanical details are illustrative; exact geometry fidelity needs continued independent review.

A01-aerial / control and finished image

A01 open-court housing / local aerial / Full build-out scenario; node core enabled in Phase 1



ORIGINAL GEOMETRY CONTROL

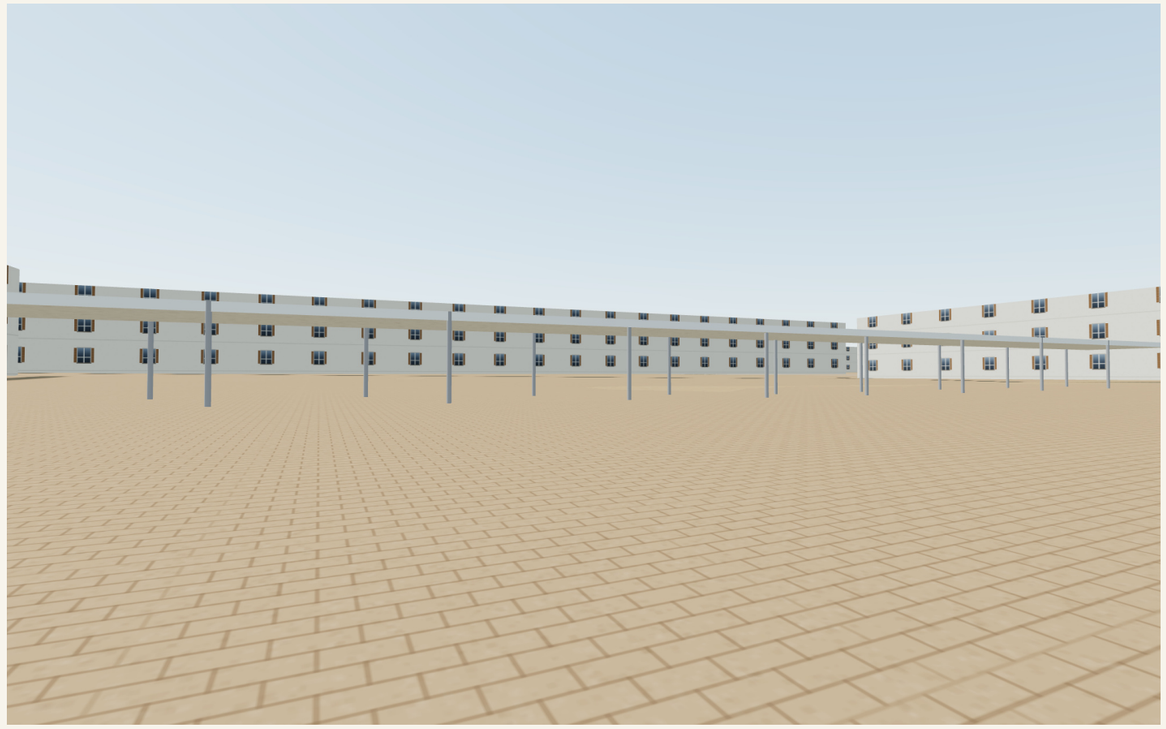


SELECTED FINISHED CONCEPT

Camera UTM19S: [574319.9999999997, 6554560.000000003, 260]; target [574099.9999999997, 6554800.000000003, 0]; FOV 46 degrees. Terrain and cameras come from the current model. Maximum-corner conceptual building plinths prevent terrain clipping and require civil design. AI materials, small people, furniture and mechanical details are illustrative; exact geometry fidelity needs continued independent review.

A01-eye / control and finished image

A01 shared court / eye-level / Full build-out scenario; node core enabled in Phase 1



ORIGINAL GEOMETRY CONTROL

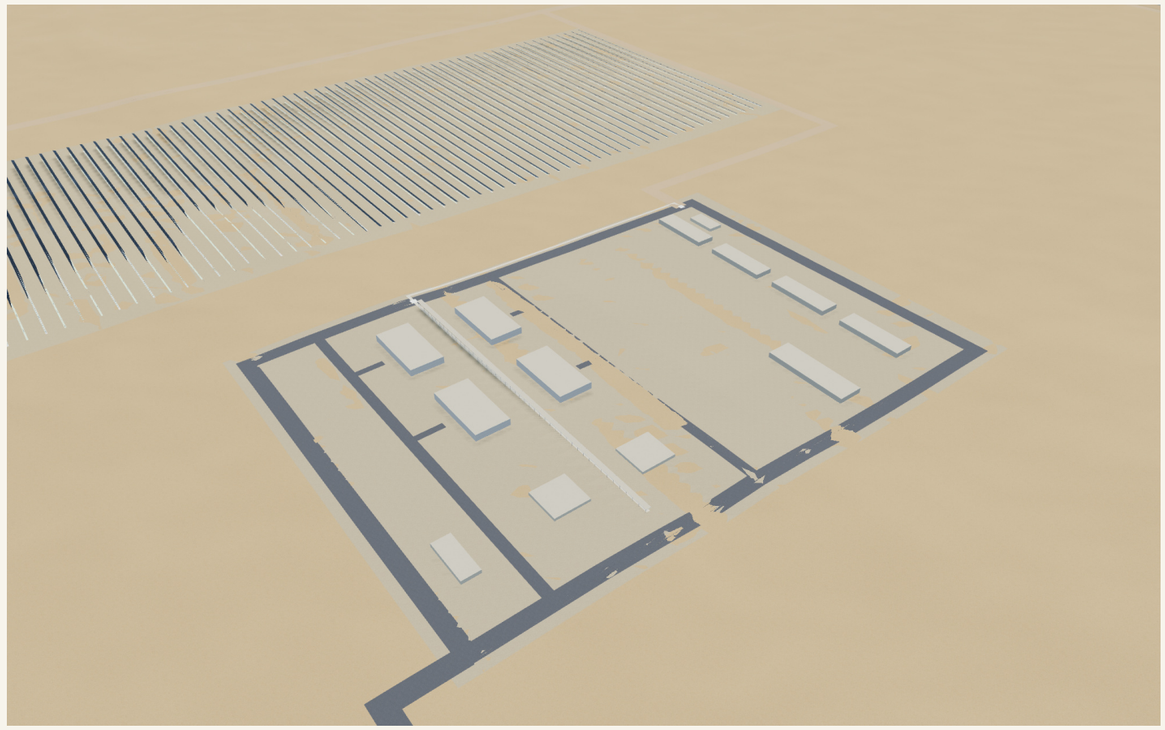


SELECTED FINISHED CONCEPT

Camera UTM19S: [574071.9999999997, 6554770.000000003, 1.7]; target [574107.9999999997, 6554850.000000003, 2]; FOV 63 degrees. Terrain and cameras come from the current model. Maximum-corner conceptual building plinths prevent terrain clipping and require civil design. AI materials, small people, furniture and mechanical details are illustrative; exact geometry fidelity needs continued independent review.

B03-aerial / control and finished image

B03 four modular industrial halls and repair / local aerial / Full build-out scenario; node core enabled in Phase 1



ORIGINAL GEOMETRY CONTROL

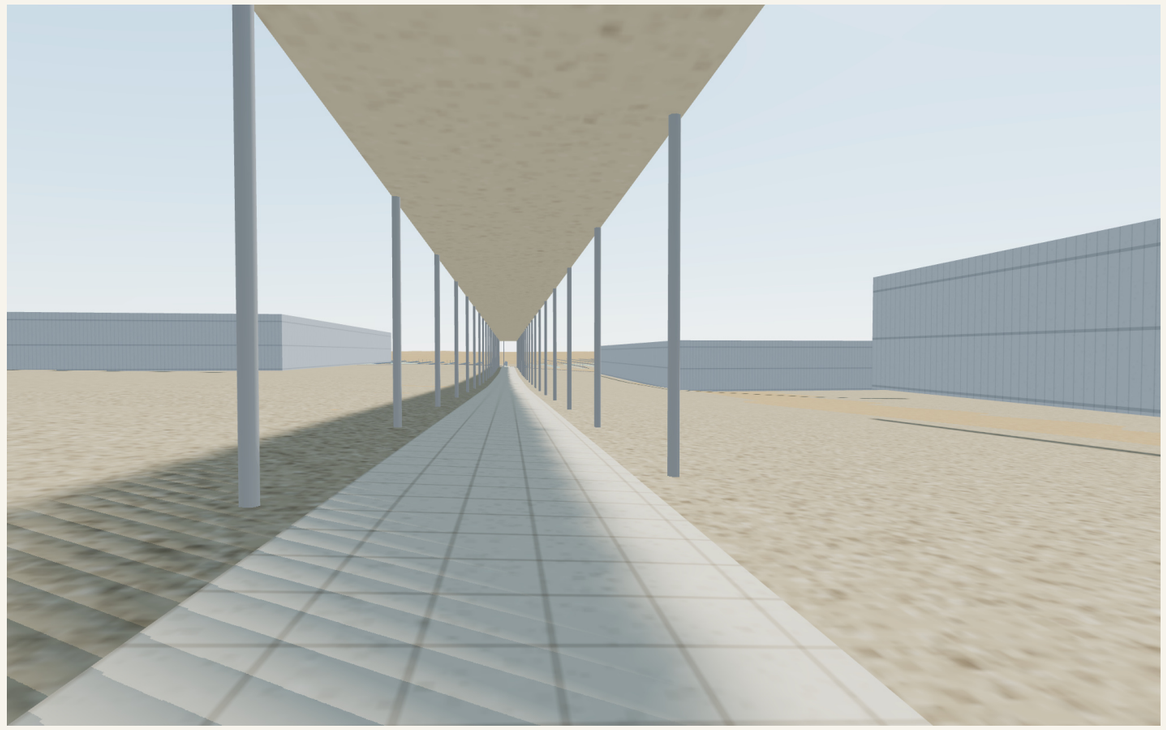


SELECTED FINISHED CONCEPT

Camera UTM19S: [574950, 6552000, 580]; target [574340, 6552460, 0]; FOV 44 degrees. Terrain and cameras come from the current model. Maximum-corner conceptual building plinths prevent terrain clipping and require civil design. AI materials, small people, furniture and mechanical details are illustrative; exact geometry fidelity needs continued independent review.

B03-eye / control and finished image

B03 protected worker spine / eye-level / Full build-out scenario; node core enabled in Phase 1



ORIGINAL GEOMETRY CONTROL

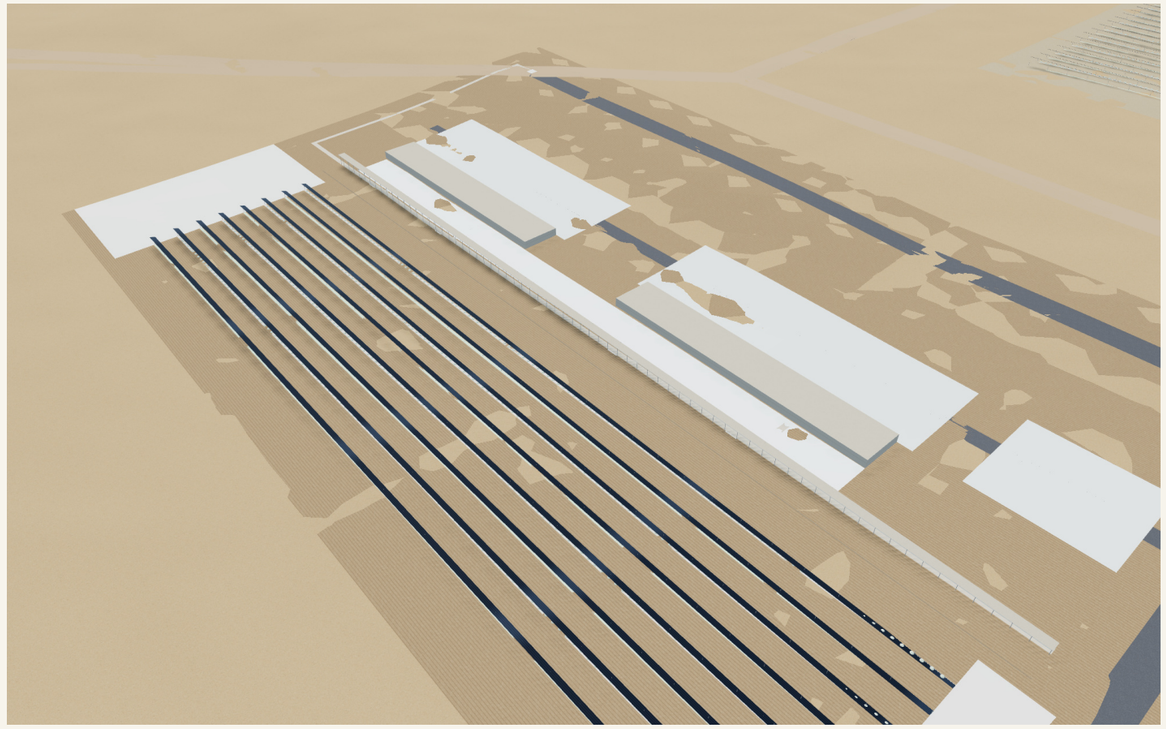


SELECTED FINISHED CONCEPT

Camera UTM19S: [574340, 6552373, 1.7]; target [574210, 6552390, 3]; FOV 64 degrees. Terrain and cameras come from the current model. Maximum-corner conceptual building plinths prevent terrain clipping and require civil design. AI materials, small people, furniture and mechanical details are illustrative; exact geometry fidelity needs continued independent review.

C02-aerial / control and finished image

C02 close local aerial / central crop rows and workshop forecourts; field ends cropped / Full build-out scenario; node core enabled in Phase 1



ORIGINAL GEOMETRY CONTROL

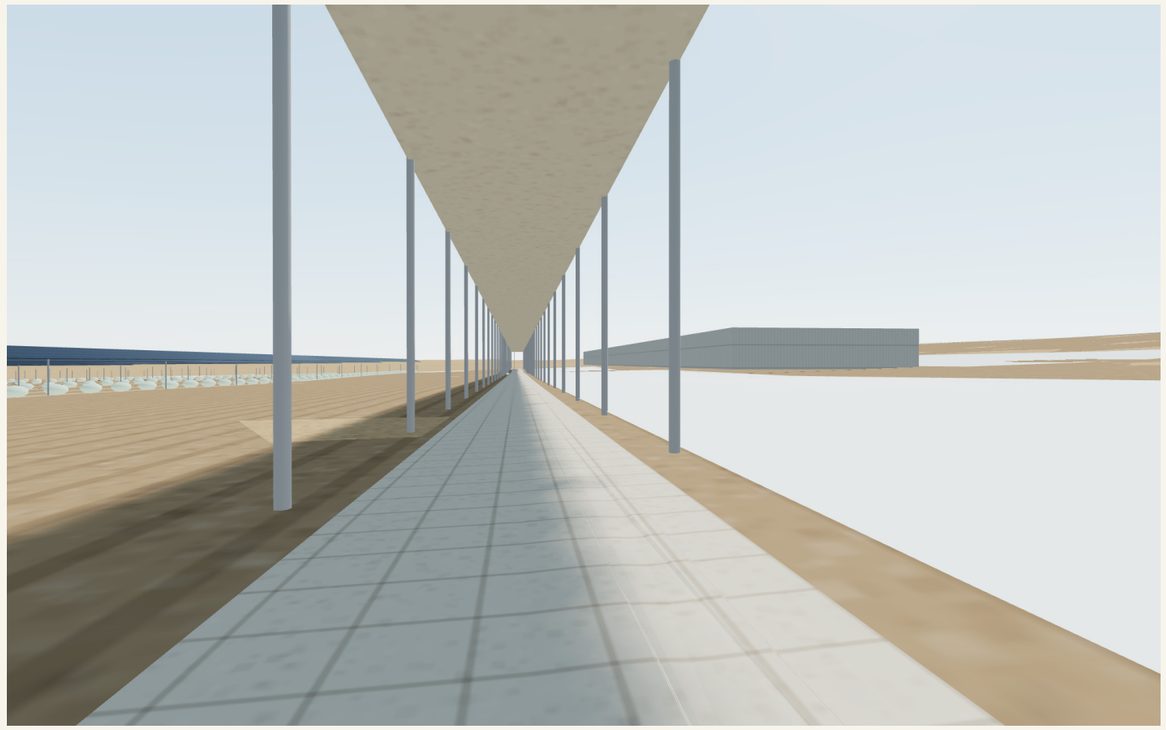


SELECTED FINISHED CONCEPT

Camera UTM19S: [573740, 6551020, 340]; target [573430, 6551290, 0]; FOV 46 degrees. Terrain and cameras come from the current model. Maximum-corner conceptual building plinths prevent terrain clipping and require civil design. AI materials, small people, furniture and mechanical details are illustrative; exact geometry fidelity needs continued independent review.

C02-eye / control and finished image

C02 occupied shade promenade and connected workshop forecourt / eye-level / Full build-out scenario; node core enabled in Phase 1



ORIGINAL GEOMETRY CONTROL



SELECTED FINISHED CONCEPT

Camera UTM19S: [573385, 6551313, 1.7]; target [573190, 6551335, 2]; FOV 63 degrees. Terrain and cameras come from the current model. Maximum-corner conceptual building plinths prevent terrain clipping and require civil design. AI materials, small people, furniture and mechanical details are illustrative; exact geometry fidelity needs continued independent review.

P1 operations and the 30 percent water stress test

Phase-one gross demand and the proposed public-service rules; all operating inputs remain declared scenarios.

P1 water use	m3/year / assumed basis
Domestic + civic	162,060 / 4,440 people x100L/day
Compute makeup	4,555.2 / 4MW IT, PUE1.3, WUE0.1L/kWh
Industry + panel cleaning	10,502.5 / worker use plus1,000 panel cleaning
New forage / jojoba irrigation	22,968 / 7.656ha explicit crop bands, screening allowance
Gross total	200,085.7 / fixed-pilot consumption additional unknown
30 percent shortfall	60,025.71 / remaining140,059.99

Water stress sequence. Halt new occupation. First curtail the assumed agricultural allocation(22,968), with crop losses and critical plant survival reviewed by an agronomist. Then stop nonessential panel cleaning(1,000), compute makeup(4,555.2) and nonessential work use(7,330.5). Halving civic use saves12,154.5; the remaining12,017.51 reduction would reduce the domestic allowance from85 to about77.6L/person/day. Essential-service allocation and recovery must be independently verified before restrictions; normal public life is not assumed to continue.

Firm obligations. The proposed public/cooperative utility owns essential interfaces; compute is a replaceable concession with a decommissioning bond. Publish daily aggregate meters, monthly accounts and a quarterly independent audit. Household traces stay private. P1 assumes an unsecured12MW grid connection;0hours storage and0% hourly clean matching are credited. Occupancy requires actual income, utility and service-funding evidence.

Source record and visual review

Public package: original model, cameras, controls, prompts, selected illustrations and dated review remain available separately.

Issued sources. Scope and fixed pilots: brief/site-package/boundaries and pilot-area. Terrain: Copernicus DEM GLO-30 source package, with its supplied licence and credits. Channels/flood bands: organiser DEM-derived studies, not surveyed limits. External roads: supplied OpenStreetMap extract; no on-site national/provincial road is asserted.

Retained original work. The earlier Everyday City terrain-cost regional corridor study and steep/high terrain study are retained with attribution to reviews/withdrawn-v1. No other entry's urban layout is copied. All town, housing, campus, farm-edge and phase geometry is new for V2.

Visual review. Eight distinct geometry cameras support two city views and aerial/eye-level pairs at A01, B03 and C02. Earlier overall photorealistic finishes altered hall counts or added planting and were rejected. The selected finished architectural-model view preserves the four-hall composition. Controls never substitute for finished public views. Canopy members, access, building gaps, four compute halls and two farm sheds have been visually compared; this is not a pixel-level or construction certification.

Remaining verification. Survey, detailed grading, noise/dust and thermal impacts, accessible routes, junction/crossing design, canopy engineering, resource rights, hourly grid/storage modelling, school operation and demand/employment evidence. Status remains an authorised public preview; automatic checks and content review are reported separately.

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